COURSE TITLE : PROCESS CONTROL LAB
COURSE CODE : 5217
COURSE CATEGORY : A
PERIODS/WEEK : 4
PERIODS/SEMESTER: 52
CREDITS : 2

On completion of this course the student will be able to

1. To design and setup Error detector circuit using OP Amplifier.
2. To design and setup Proportional controller circuit using OP Amplifier.
3. To design and setup Proportional Derivative controller circuit using OP Amplifier.
4. To design and setup Proportional integral controller circuit using OP Amplifier.
5. To design and setup PID controller circuit using OP Amplifier.
6. To design and setup ON/OFF Controller using OP Amplifier.
7. To design and setup Instrumentation Amplifier using OP Amplifier.
8. To design and setup Current To voltage converter circuit.
9. To design and Setup Voltage to Current converter circuit.
10. To plot the characteristic of Level Transmitter.
11. To plot the characteristic of Temperature Transmitter.
12. To plot the characteristic of Pressure Transmitter.
13. To plot the characteristic of Flow Transmitter.
14. To plot the curve between stem position Vs flow rate of a control valve.
15. To plot the curve between actuator Vs flow rate of a control valve.
16. To find the flow coefficient of control valve.

Practices using MATLAB and SIMULINK

17. To simulate Proportional controller using MATLAB and SIMULINK.
18. To simulate PI controller using MATLAB and SIMULINK.
19. To simulate PD controller using MATLAB and SIMULINK.
20. To simulate PID controller using MATLAB and SIMULINK.
21. To simulate Instrumentation Amplifier using MATLAB and SIMULINK.